Exercise Sheet 3: Determinants

October 2024

In Exercises 1-4, find the determinant of the matrix.

1. $\begin{bmatrix} 1 & 4 & -2 \\ 0 & -3 & 1 \\ 1 & 1 & -1 \end{bmatrix}$.

2. $\begin{bmatrix} 2 & 0 & -1 & 4 \\ -1 & 2 & 0 & 3 \\ 3 & 0 & 1 & 2 \\ -2 & 0 & 3 & 1 \end{bmatrix}$.

3. $\begin{bmatrix} -4 & 1 & 2 & 3 \\ 1 & -2 & 1 & 2 \\ 2 & -1 & 3 & 4 \\ 1 & 2 & 2 & -1 \end{bmatrix}$.

4. $\begin{bmatrix} -1 & 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 \\ 1 & 0 & 1 & -1 & 0 \\ 0 & -1 & 0 & 1 & -1 \\ 0 & 1 & 1 & -1 & 1 \end{bmatrix}$.

5. Find (a) $|A^T|$, (b) $|A^3|$, (c) $|A^T A|$, (d) $|A^{-1}|$ and (e) $|5A|$, where

$$A = \begin{bmatrix} 1 & 0 & -4 \\ 0 & 3 & 2 \\ -2 & 7 & 6 \end{bmatrix}.$$

In Exercices 6-7, use Cramer's Rule to solve the system of linear equations.

6. $\begin{aligned} x_1 + 2x_2 &= 5 \\ -x_1 + x_2 &= 1. \end{aligned}$

7. $\begin{aligned} 2x_1 + x_2 + 2x_3 &= 6 \\ -x_1 + 2x_2 - 3x_3 &= 0 \\ 3x_1 + 2x_2 - x_3 &= 6. \end{aligned}$

In Exercises 8-9, use the determinant of the coefficient matrix to determine whether the system has a unique solution.

8. $\begin{aligned} -x + y + 2z &= 1 \\ 2x + 3y + z &= -2 \\ 5x + 4y + 2x &= 4. \end{aligned}$

9. $\begin{aligned} x_1 + 2x_2 + 6x_3 &= 1 \\ 2x_1 + 5x_2 + 15x_3 &= 4 \\ 3x_1 + x_2 + 3x_3 &= -6. \end{aligned}$

In Exercises 10-11, find the adjoint of the matrix A, then use the adjoint to find the inverse of A.

10. $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -1 \\ 2 & 2 & 2 \end{bmatrix}$.

11. $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 2 & 3 \\ -1 & -1 & -2 \end{bmatrix}$.

12. Find the eigenvalues and corresponding eigenvectors of the matrix $A = \begin{bmatrix} -3 & 10 \\ 5 & 2 \end{bmatrix}$.